Spectral-Spatial and Superpixelwise PCA for Unsupervised Feature Extraction of Hyperspectral Imagery

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IE506 Course Project

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  - Dimensionality Reduction using PCA
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- Proposed method - S^3-PCA
- Details of Experiments
- Experimental results
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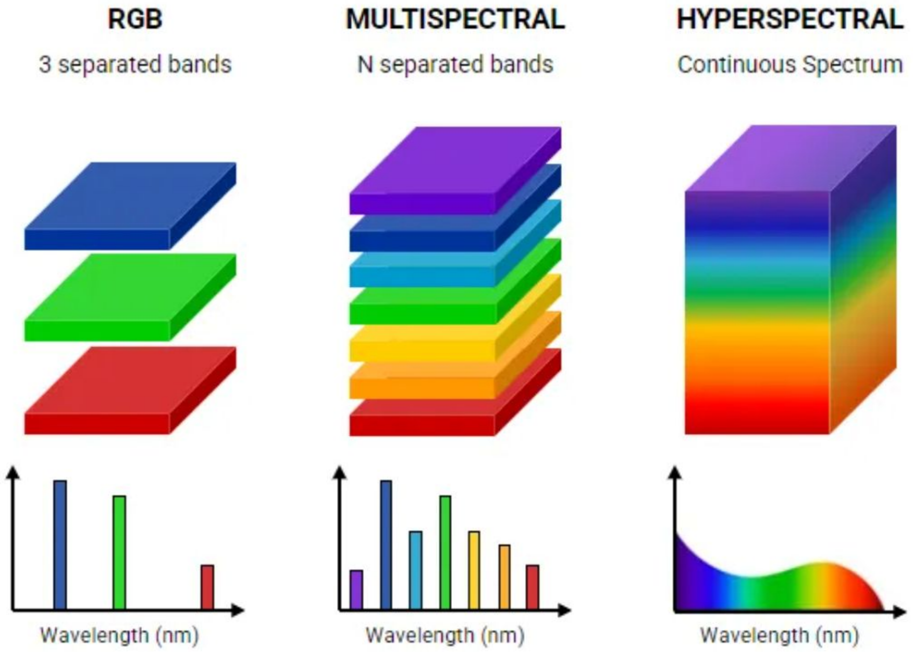
Introduction - Hyperspectral Imagery

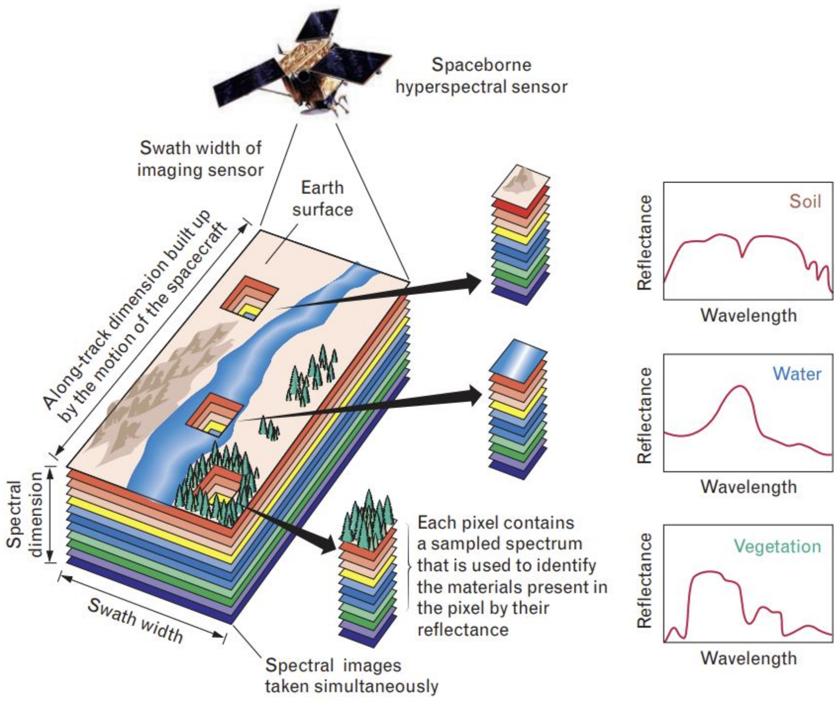
- RGB images capture intensity in three broad separated spectral bands (Red, Green, Blue).

- Hyperspectral images (HSIs) record hundreds of narrow continuous spectral bands, providing detailed spectral information.



Nireos. *What is Hyperspectral Imaging?* [Online]. Available: https://nireos.com/application/what-is-hyperspectral-imaging/. Accessed on: 18 March 2025.

Introduction - Hyperspectral Imagery



EO Portal. *Hyperspectral Imaging*. [Online]. Available: https://www.eoportal.org/other-space-activities/hyperspectral-imaging. Accessed on: 18 March 2025.

- HSI is used in remote sensing applications. Each material has a unique spectral signature. The reflectance properties captured in HSIs can be used to classify materials.

- However, due to the high dimensionality of the data, the problem of curse of dimensionality occurs, which reduces generalization capability of classifiers. To alleviate this problem, dimensionality reduction (DR) methods are used as a preprocessing step.

Introduction - DR using PCA

- Principal Component Analysis (PCA) is the most widely used unsupervised DR algorithm. It aims to find the optimal linear projection of high-dimensional data to a low-dimensional subspace by maximising the variance in each projected dimension.

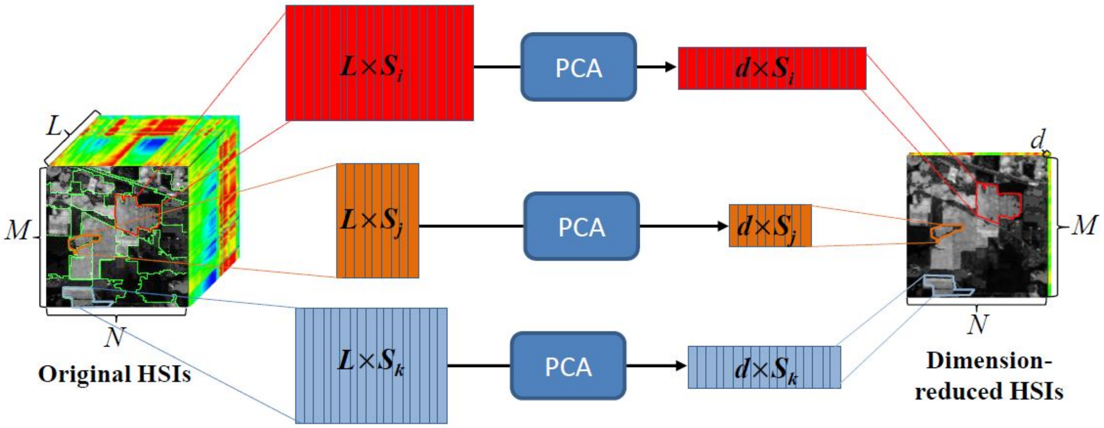
- Mathematically, the transformation between the data matrix $\mathbf{X}$ and its projection $\mathbf{Y}$ can be expressed as $\mathbf{Y} = \mathbf{W}^T\mathbf{X}$. The objective function of PCA is $\mathbf{W}^* = \text{argmax}\ \text{Tr}(\mathbf{W}^T\text{Cov}(\mathbf{X})\mathbf{W})$ subject to the constraint $\mathbf{W}^T\mathbf{W} = \mathbf{I}$, where $\text{Cov}(\mathbf{X})$ is the covariance matrix of $\mathbf{X}$.

- PCA can be used for DR in HSIs but performs poorly when used as a preprocessing step for classification. It considers only spectral information, ignoring spatial correlations. Since neighboring pixels are likely of the same class, incorporating spatial information improves feature representation and classification accuracy.

Previous works - SuperPCA

- *Jiang et al.* [1] reasoned that different ‘homogeneous’ regions of HSIs will have different optimal projection matrices.

- They proposed the Superpixelwise PCA (SuperPCA) method, that involves segmenting the image into several homogeneous regions, and obtaining reduced spectral data for each region.



Jiang, J., Ma, J., Chen, C., Wang, Z., Cai, Z., & Wang, L. *SuperPCA: A Superpixelwise PCA Approach for Unsupervised Feature Extraction of Hyperspectral Imagery*. IEEE Transactions on Geoscience and Remote Sensing, 56(8), 4581-4593, 2018.

Previous works - SuperPCA

Steps involved in SuperPCA -

1. Perform PCA on the HSI data and project it onto the first principal component.

2. Use this projection to perform image segmentation using superpixelization algorithms like Entropy Rate Superpixel Segmentation (ERS) [2] or Simple Linear Iterative Clustering (SLIC) [3].

3. Perform PCA in each superpixel separately, and project the HSI data onto the first few principal components.

4. Combine these low-dimensional matrices to form the dimensionality reduced HSI.

Previous works - SuperPCA

Limitations -

- SuperPCA only considers local spatial information and ignores the global structures of the HSI.
- This can compromise classification accuracy if there is limited labelled data in small homogenous regions, or mixed ground truth objects (pixels with different labels) in large superpixel blocks.
- These limitations are addressed in the S^3-PCA method.

Proposed method - S^3-PCA

- *Zhang et al.* [4] proposed the Spectral-Spatial and Superpixelwise PCA (S^3-PCA) method to overcome the limitations of SuperPCA and improve classification accuracy.

- There are two major modifications in this method -

    - A superpixels-based local reconstruction of the HSI is performed to eliminate noise.

    - PCA is performed on the global HSI and each superpixel block, and the global and local features are concatenated to form a global-local and spectral-spatial feature set.

Proposed method - S^3-PCA

Superpixel Segmentation using ERS (Used by Authors) -

- Entropy Rate Superpixel Segmentation (ERS) is a graph-based superpixel segmentation algorithm that ensures compact, homogeneous clusters.

- In ERS, the superpixel segmentation is formulated as a graph partitioning problem.

- Given a graph **G** = (**V**, **E)**, where **V** and **E** represent the pixels and the associativity between the pixels respectively, the goal is to partition **G** into **K** connected subgraphs by selecting a subset of edges **A**, while maximising the objective function

$$Tr\,\{H(\mathbf{A}) + \alpha B(\mathbf{A})\}, \quad s.t.\ \mathbf{A} \subseteq \mathbf{E}$$

  Here, $H(\mathbf{A})$ is the entropy rate term for generating homogeneous clusters, while $B(\mathbf{A})$ is a balancing term to encourage clusters with similar sizes.

- The combinatorial graph partitioning problem is solved using a greedy algorithm.

Proposed method - S³-PCA

Superpixel Segmentation using SLIC (Used for this project) -

- Simple Linear Iterative Clustering (SLIC) is a clustering-based superpixel segmentation method that groups pixels into compact and homogeneous regions in a 5D feature space (color (*lab*) + spatial position (*xy*)).

- The distance measure $\boldsymbol{D_s}$ is defined as $D_s = d_{lab} + \frac{m}{S} d_{xy}$ where

$$d_{lab} = \sqrt{(l_k - l_i)^2 + (a_k - a_i)^2 + (b_k - b_i)^2} \qquad d_{xy} = \sqrt{(x_k - x_i)^2 + (y_k - y_i)^2}.$$

S is the mean superpixel size and *m* is a parameter which controls the compactness (spatial proximity) of the clusters.

- *K-means* clustering is performed in the 5D feature space (3D for grayscale images) using this distance measure.

Proposed method - S^3-PCA

Superpixel-based local reconstruction -

- Each pixel is reconstructed using a weighted sum of its k nearest pixels within the superpixel, to eliminate noise and sharpen spectral features in the HSI.

- For each sample $\mathbf{x}_i$, its k nearest pixels in the superpixel block are denoted by $\mathbf{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_k\}$.

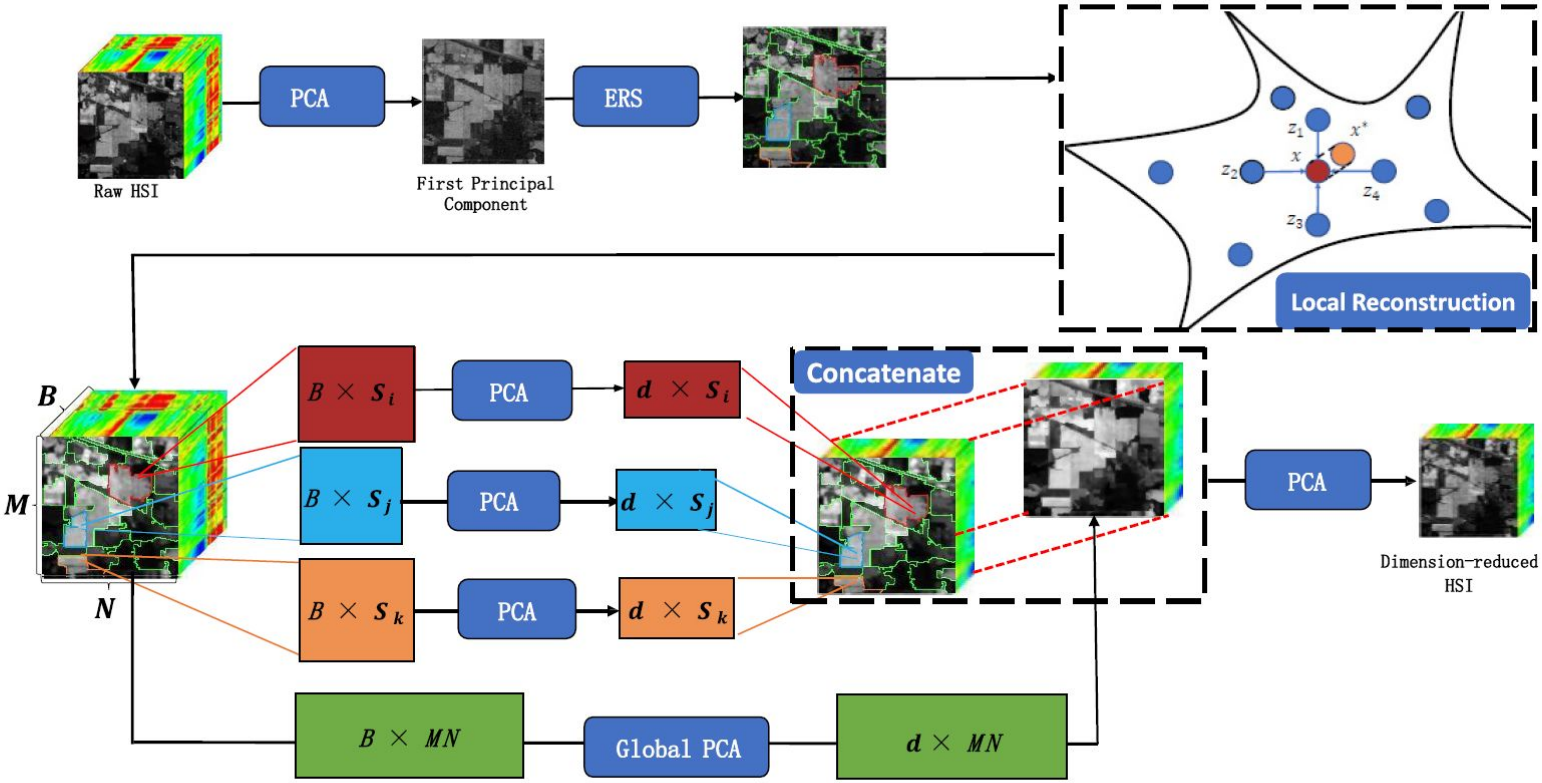
- The similarity weight between $\mathbf{x}_i$ and $\mathbf{z}_j$ is given by

$$w_{ij} = \frac{exp(-||\mathbf{x}_i - \mathbf{z}_j||_2^2/2t^2)}{h}$$

where $t^2 = \frac{1}{k}\Sigma_{j=1}^{k}||\mathbf{x}_i - \mathbf{z}_j||_2^2$, $h = \frac{1}{k}\Sigma_{j=1}^{k} exp(-||\mathbf{x}_i - \mathbf{z}_j||_2^2/2t^2)$.

- The reconstructed pixel is computed as $\mathbf{x}_i^* = \Sigma_{j=1}^{k} w_{ij}\mathbf{z}_j$.

Proposed method - S³-PCA



Zhang, X., Jiang, X., Jiang, J., Zhang, Y., Liu, X., & Cai, Z. *Spectral-Spatial and Superpixelwise PCA for Unsupervised Feature Extraction of Hyperspectral Imagery*. IEEE Transactions on Geoscience and Remote Sensing, 60, 1-10, 2022.

Implementation by Authors

- Superpixel Segmentation: ERS algorithm implemented in C++ by *Liu et al.* (GitHub - mingyuliutw/EntropyRateSuperpixel: Entropy rate superpixel segmentation source code), called from MATLAB.

- Feature Extraction: SuperPCA & S3-PCA implemented in MATLAB.

- SVM Classification: LibSVM (C++ library) (cjlin1/libsvm - A Library for Support Vector Machines) used for classification, called from MATLAB.

- Programming Languages: MATLAB (dimensionality reduction) and C++ (superpixelization, LibSVM).

- Code Repository: https://github.com/junjun-jiang/SuperPCA/tree/master
  https://github.com/XinweiJiang/S3-PCA/tree/master

Details of Experiments

Datasets -

1. Indian Pines Scene :
- Spatial dimensions : 145 × 145 pixels
- Spectral bands : 224 bands covering wavelengths from 0.4 to 2.5 micrometers.
- Classes : 16 land cover types.

2. Pavia University Scene :
- Spatial Dimensions: 610 × 340 pixels.
- Spectral Bands: 115 bands in the 0.43 to 0.86 micrometers range.
- Classes: 9 urban land cover types.

Both datasets are publicly available at Hyperspectral Remote Sensing Scenes - Grupo de Inteligencia Computacional (GIC).

Details of Experiments

- Dimensionality: The feature dimension is reduced to 30 for all experiments.

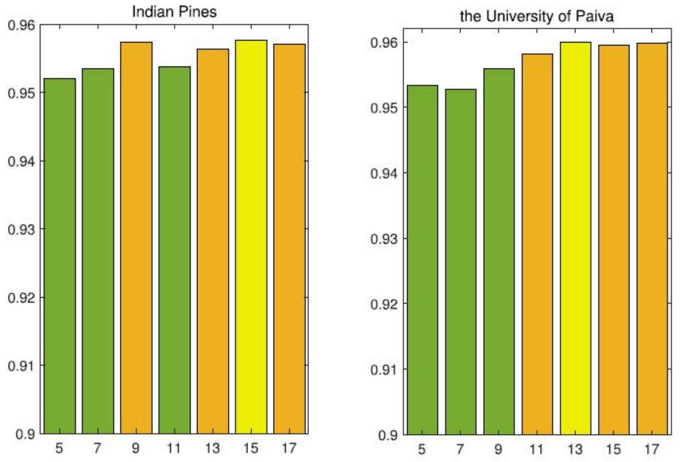
- Training Samples: T = 5, 10, 20, 30, 40, 50, 60 samples per class are randomly selected from the HSI.

- Classifier: SVM with RBF kernel, with hyperparameters optimized via cross-validation.

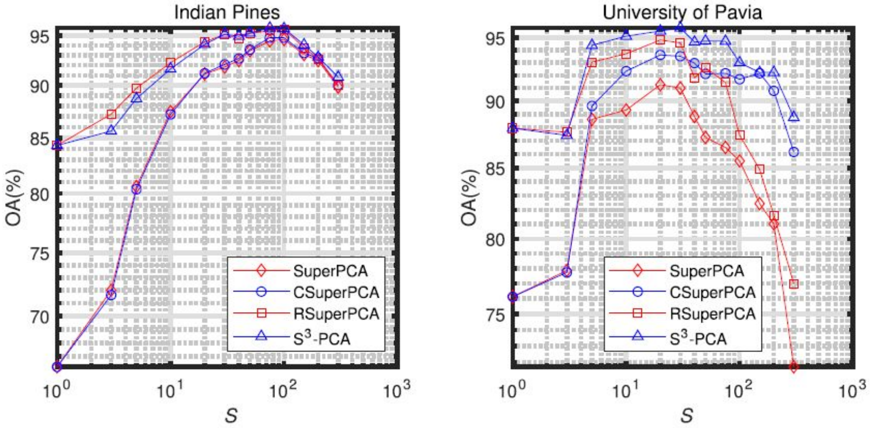
- S^3-PCA Parameters: Optimal values for number of segments in superpixelization (S) and number of neighbors used for reconstruction (k) are found using experiments.

- Benchmarking: After dimensionality reduction, the optimal trained SVM is used to classify test samples, and results are compared against state-of-the-art DR methods.

Experimental Results



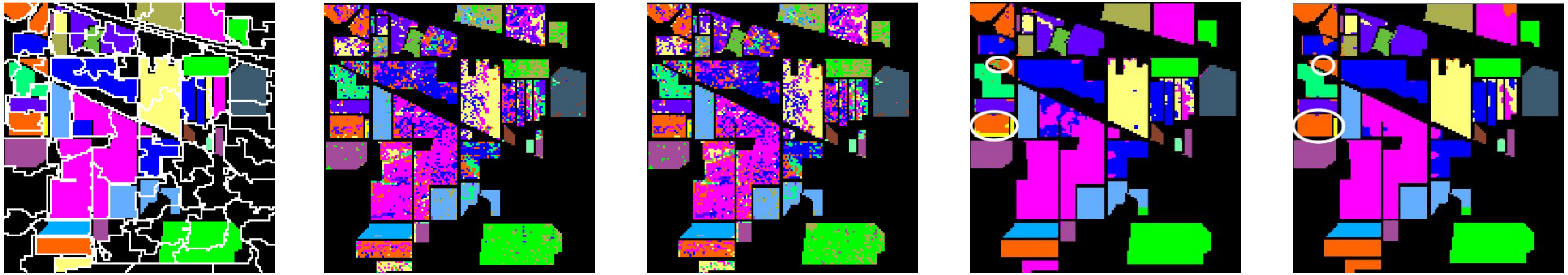
OAs for different values of *k* when *S* is set to 100 and 20 respectively. *k* = 15, 13 are found to be optimal.



OAs for different values of *S* when *k* is set to 15 and 13 respectively. *S* = 75, 30 are found to be optimal.

Zhang, X., Jiang, X., Jiang, J., Zhang, Y., Liu, X., & Cai, Z. *Spectral-Spatial and Superpixelwise PCA for Unsupervised Feature Extraction of Hyperspectral Imagery*. IEEE Transactions on Geoscience and Remote Sensing, 60, 1-10, 2022.

Experimental Results



Ground Truth | Raw data | PCA | SuperPCA | S^3-PCA

Classification maps obtained using different methods for preprocessing the Indian Pines scene. (Number of training samples per class = 30)

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Experimental Results

Overall Accuracies from different methods for the two HSI datasets with different number of training samples

Datasets	T.N.s/C	Raw	PCA	LPP	NPE	LNSPE	LPNPE	NWFE	LFDA	SuperPCA	S^3-PCA
Indian Pines	5	44.87%±6.50%	45.47%±5.33%	51.56%±6.89%	53.26%±6.42%	55.01%±4.48%	71.73%±3.66%	60.08%±2.68%	59.12%±3.75%	77.14%±5.16%	**80.36%±4.21%**
	10	55.77%±3.27%	55.07%±2.87%	71.09%±2.60%	71.44%±3.39%	73.73%±2.94%	81.60%±3.87%	70.03%±2.52%	64.06%±2.95%	85.75%±3.06%	**87.44%±1.37%**
	20	63.81%±3.37%	62.16%±2.64%	81.88%±1.49%	82.84%±1.37%	85.28%±1.42%	89.41%±1.73%	85.08%±1.67%	82.48%±2.02%	92.80%±1.46%	**94.16%±1.61%**
	30	68.77%±1.27%	66.24%±0.58%	85.45%±2.16%	86.45%±1.91%	88.78%±1.91%	92.74%±1.56%	88.86%±1.61%	87.58%±1.14%	94.61%±0.81%	**95.81%±0.77%**
	40	71.64%±1.05%	68.90%±1.06%	87.51%±1.68%	88.54%±1.32%	90.60%±1.19%	93.37%±1.56%	90.37%±1.48%	89.68%±0.60%	95.33%±0.97%	**96.24%±0.84%**
	50	74.18%±1.20%	70.70%±1.01%	89.15%±1.20%	89.95%±1.26%	91.96%±1.15%	94.57%±1.26%	91.78%±1.32%	90.97%±0.75%	95.42%±1.02%	**96.67%±0.85%**
	60	75.24%±1.12%	71.79%±1.33%	90.16%±0.92%	90.97%±0.72%	93.01%±0.85%	95.25%±0.89%	92.52%±0.81%	91.71%±0.91%	95.72%±0.56%	**97.05%±0.85%**
University of Pavia	5	64.59%±5.07%	65.26%±5.14%	70.62%±5.79%	68.66%±6.05%	68.26%±8.56%	79.63%±3.17%	72.18%±5.46%	75.27%±2.83%	74.36%±3.65%	**83.70%±3.41%**
	10	70.22%±3.05%	67.00%±3.06%	79.12%±3.82%	77.76%±4.24%	78.71%±6.26%	86.09%±2.72%	81.31%±2.27%	78.90%±2.84%	83.39%±3.33%	**90.88%±2.58%**
	20	75.85%±2.37%	75.80%±2.31%	86.92%±1.60%	84.70%±1.91%	87.71%±2.02%	90.62%±1.89%	89.14%±1.94%	81.65%±2.14%	89.34%±1.54%	**95.63%±1.10%**
	30	76.45%±1.50%	76.13%±1.85%	89.59%±1.64%	87.33%±1.69%	90.61%±1.41%	91.70%±1.12%	90.64%±1.60%	85.56%±1.62%	91.26%±0.99%	**96.24%±1.33%**
	40	77.76%±1.26%	77.41%±1.49%	91.12%±1.45%	89.13%±1.17%	92.50%±1.08%	93.35%±0.81%	91.77%±1.19%	88.27%±1.33%	92.19%±0.81%	**96.72%±1.14%**
	50	79.12%±1.31%	78.77%±1.31%	92.38%±1.23%	91.01%±1.46%	93.62%±0.79%	94.09%±0.68%	92.66%±0.65%	89.75%±1.58%	93.25%±0.90%	**97.34%±0.77%**
	60	80.52%±1.32%	79.87%±1.16%	93.52%±0.89%	92.05%±1.07%	94.59%±0.62%	94.74%±0.80%	93.62%±0.92%	91.40%±1.13%	93.99%±1.08%	**97.84%±0.49%**

- S^3-PCA outperforms all unsupervised and supervised dimensionality reduction methods considered in the study.
- The accuracy improvement is most significant when fewer training samples are available.
- The strong performance with limited training data suggests S^3-PCA is most beneficial for sparsely labelled HSIs.

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Status of Work

Tasks completed -

- Python implementation of SuperPCA method using SLIC (available in *scikit-image*) for image segmentation.
- Python implementation of S^3-PCA method using SLIC.
- Performed SVM classification (using *scikit-learn*) on one of the datasets using both methods.

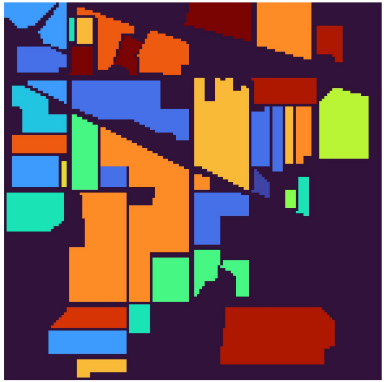
Tasks to be done -

- Experiments to assess the impact of the various parameters (number of superpixels, number of nearest neighbors used in reconstruction, number of principal components) on different datasets.
- Implementation of the graph-based ERS segmentation algorithm (if time permits).

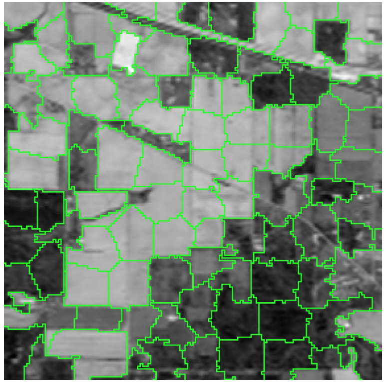
Preliminary Results

Dataset - Indian Pines scene [7]

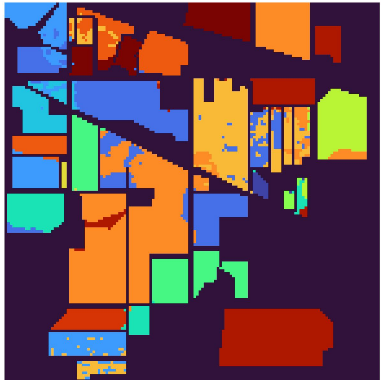
Number of segments - 97
Number of neighbours used for reconstruction (S^3-PCA) - 15
Number of training samples per class - 30



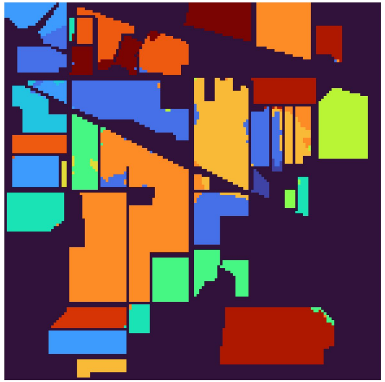
Ground Truth



Segmented Image
(First Principal Component)



SuperPCA
Accuracy - 91.2 %



S^3-PCA
Accuracy - 95.4 %

References

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3. R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC Superpixels Compared to State-of-the-Art Superpixel Methods," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 34, no. 11, pp. 2274–2282, 2012. doi: 10.1109/TPAMI.2012.120.

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Use of AI tools

AI Tools Used : ChatGPT (GPT-4)

Tasked AI tools for :

- Function usage for superpixel segmentation using skimage.
- Syntax for implementation of SVM for classification using scikit-learn.
- Enhancing readability and clarity of presentation slides.